

DIALLO BOYE

Senior Data Scientist | ML & Forecasting

+33 6 45 06 40 47
diallo.boye@cern.ch
Saint Genis, FRANCE



PROFILE

Senior data scientist with 6+ years at CERN and Brookhaven National Laboratory, specialising in statistical modeling, time series analysis, and end-to-end ML pipeline development on large-scale, high-dimensional datasets. Experienced in owning projects from problem framing to production, translating complex quantitative findings into actionable insights, and leading cross-functional teams. Author of multiple peer-reviewed publications in data-driven analysis and predictive modeling.

CORE COMPETENCIES

Time Series & Forecasting Temporal modeling, trend/seasonality decomposition, backtesting, cross-validation	Machine Learning Classification, regression, XGBoost/LightGBM, neural networks, feature engineering	Statistical Modeling Bayesian & frequentist inference, uncertainty quantification, experimental design
Data Engineering Petabyte-scale pipelines, distributed computing, real-time processing, SQL	Programming Python (NumPy, pandas, scikit-learn, SciPy, PyTorch), C++, SQL, Linux/bash	Communication & Leadership Cross-functional team lead, stakeholder reporting, 6+ international conference talks

PROFESSIONAL EXPERIENCE

Goldhaber Distinguished Fellow - Brookhaven National Laboratory / CERN 2023 – 2026

Highly competitive fellowship awarded to fewer than 5 early-career scientists per year worldwide.

- ▶ Developed time series analysis pipelines for beam parameter evolution — extracting trends, modeling non-linear dynamics, and delivering sub-percent precision results used across an international collaboration of 3,000+ scientists.
- ▶ Led a cross-functional team of ~20 researchers, setting technical direction, coordinating deliverables, and mentoring junior scientists across multiple concurrent projects.
- ▶ Designed and deployed automated ML pipelines replacing thousands of hand-tuned fits with a unified, scalable model — improving robustness and precision in production.
- ▶ Invited plenary speaker at LHCP 2026 (Paris) — selected to present a cross-experiment review to 500+ scientists; 6 further international conference talks.

Postdoctoral Research Associate - Brookhaven National Laboratory 2020 – 2023

- ▶ Built end-to-end ML pipelines on petabyte-scale datasets: feature engineering, model training, cross-validation, backtesting, and production monitoring — with results published in peer-reviewed journals.
- ▶ Applied multivariate classifiers and statistical inference to extract rare signals from overwhelming backgrounds — directly analogous to demand forecasting in volatile, noisy environments.
- ▶ Mentored 4 graduate students across 3 continents in quantitative methods, Python, and scientific software development.

PhD Researcher - CERN / ATLAS 2016 – 2020

- ▶ Implemented full data science workflows: simulation, classification, anomaly detection, hypothesis testing, and automated reporting on multi-terabyte datasets.
- ▶ Developed real-time statistical monitoring tools for data quality assurance, integrating live data streams with automated alerting.

Research Intern - CERN Feb – Jul 2016

- ▶ Contributed to a precision measurement requiring sub-percent accuracy through rigorous statistical modeling of high-dimensional data.

EDUCATION

PhD, Particle Physics - ATLAS / CERN / University of South Africa	2020
MSc, Physics of Energy & Energy Transition - Universite Paul Sabatier, Toulouse	2016
MSc, Subatomic Physics & Astroparticle - Universite de Strasbourg	2014

LANGUAGES

French Native English Fluent